Digi Pharm

CHI Sandbox

Value-Based Contracting Testing Plan

VALUE-BASED CONTRACTING TESTING PLAN

Comprehensive Testing Objectives

DigiPharm will conduct a rigorous sandbox evaluation to validate REIMBURSE as a transformative platform for healthcare procurement in the Kingdom of Saudi Arabia, addressing critical technological, contractual, and systemic challenges while generating valuable policy development insights.

Policy Development and Strategic Context

Beyond technological validation, our sandbox testing serves a critical function for the Council of Health Insurance (CHI) by:

- Exploring feasibility of value-based payment frameworks
- Identifying practical challenges in outcome-linked contracts
- Mapping systemic barriers to healthcare procurement transformation
- Generating actionable insights for policy development
- Supporting Saudi Vision 2030 healthcare transformation goals

Detailed Testing Methodology and Use Cases

Use Case 1: Medical Technology Procurement Scenario

Objective: Validate platform's ability to create dynamic, outcome-linked contracts for medical technology suppliers

Functionalities and Features to be Tested

1. Technological Infrastructure Validation

- Detailed integration with Saudi Billing System (SBS) and payment/reimbursement/rebate workflows

- Comprehensive alignment with Australian Refined Diagnosis Related Group (AR-DRG) standards
- Robust data transfer and information governance protocols

Technical Depth Includes:

- Granular mapping of medical procedure codes
- Cross-referencing existing classification systems
- Adaptive algorithms for medical coding interpretation
- Complex data scenario simulations testing:
    - Transfer latency under high-volume conditions
    - Encryption and decryption processes
    - Anonymisation without losing critical context

2. Contractual Dynamics Testing

- Dynamic, outcome-linked contract generation
- Sophisticated performance-based supplier selection
- Multi-dimensional evaluation mechanisms

Innovative Contract Approaches:

- Holistic supplier scoring systems
- Qualitative and quantitative performance indicators
- Dynamic weighting mechanisms for medical technologies
- Risk-sharing contract structures including:
    - Partial reimbursement for partially successful interventions
    - Performance-linked payment cascades
    - Adaptive pricing models based on real-world outcomes

Key Steps for Medical Technology Procurement

1. Initial Setup

- Populate platform with synthetic supplier and healthcare provider datasets
- Configure performance metrics specific to medical technology evaluation
- Establish baseline performance benchmarks

2. Contract Simulation Process

- Generate multiple contract scenarios for different medical technologies
- Create varying performance criteria
- Implement risk-sharing mechanisms
- Simulate negotiation processes between providers and suppliers

Validation Metrics for Medical Technology Use Case:

- Contract generation accuracy
- Complexity of risk allocation
- Performance tracking precision
- Cost savings potential
- Negotiation efficiency

Use Case 2: Healthcare Provider-Insurer Contracting

Objective: Test platform's capability to create transparent, outcome-driven contracts

Key Steps:

1. Data Preparation
- Develop synthetic healthcare intervention datasets
- Create anonymised performance indicators
- Configure payment linked to specific health outcomes
2. Contract Dynamics Testing
- Simulate multiple intervention scenarios
- Test adaptive pricing mechanisms
- Validate performance-linked reimbursement structures

Validation Metrics:

- Pricing mechanism flexibility
- Outcome correlation accuracy
- Administrative efficiency
- Transparency of contract terms

Use Case 3: Systemic Workflow Integration

Objective: Assess platform's interoperability with existing healthcare systems

Key Steps:

1. System Compatibility Testing

- Integration with Saudi Billing System (SBS)
- Alignment with AR-DRG classification standards
- Data inter-operability and transfer protocol validation

2. Workflow Simulation

- Test data anonymisation processes
- Validate information governance mechanisms
- Simulate complex multi-stakeholder interactions

Validation Metrics:

- System interoperability
- Data transfer accuracy
- Compliance with regulatory standards
- Workflow efficiency improvements

Success Criteria and Measurable Outcomes

Quantitative Metrics:

- 90%+ accuracy in outcome-based contract generation
- Demonstrate potential cost savings of 15-20%
- Verify 100% data anonymisation compliance
- Achieve 85% stakeholder satisfaction

Qualitative Policy Insights:

- Detailed mapping of procurement transformation challenges

- Identification of regulatory and operational barriers
- Recommendations for healthcare system optimisation

Testing Methodology Phases

Phase 1: Platform Configuration and Preparation

- Configure REIMBURSE with CHI/Saudi Market-specified parameters
- Integrate synthetic healthcare performance datasets
- Establish robust data protection protocols

Phase 2: Contractual Scenario Simulation

- Generate multiple value-based contract scenarios
- Test adaptive pricing and performance-linked payment mechanisms
- Simulate complex negotiations between providers, insurers, and suppliers

Phase 3: Comprehensive Performance and Policy Validation

- Execute detailed simulation scenarios
- Analyse contract performance metrics
- Generate comprehensive reporting on system improvements
- Develop policy recommendations

Comprehensive Validation Framework

Technical Performance Metrics:

- Data transfer reliability (99.9% accuracy target)
- System response time
- Data anonymisation effectiveness
- Encryption robustness

Contractual Performance Metrics:

- Contract generation speed
- Pricing mechanism adaptability
- Risk allocation precision
- Performance tracking accuracy

Economic Impact Metrics:

- Potential cost savings
- Efficiency improvements
- Value creation potential

Strategic Alignment and Policy Contribution

The testing directly supports CHI's objectives by:

- Transitioning from volume-based to value-based healthcare
- Enhancing procurement transparency
- Creating aligned stakeholder incentives
- Providing empirical data for policy refinement
- Supporting healthcare system innovation

Broader Healthcare Ecosystem Implications

Our sandbox testing represents a strategic exploration demonstrating how advanced procurement platforms can:

- Align stakeholder incentives
- Reduce administrative overhead
- Enhance patient-centric care delivery
- Generate actionable policy insights
- Support national healthcare transformation goals